



Fig. 2: Robust Design flow

design methods, such as Taguchi and Design For Six Sigma (DFSS), and optimise their mechatronic systems for quality and cost.

Saber links to transistor-level CAD (TCAD) tools from Synopsys enable engineering teams to achieve faster product design for power electronics systems. By abstracting the device-level physics to physical systems, designers can work with accurate, compact models of power components. These abstracted models also support behavioural circuit simulation that can be hundreds of thousands of times faster than device-level mixed-mode simulation.

Vehicle power network. As electronic content in vehicles increases, the challenges in meeting this rising energy demand on the power network also rise. Designers must create efficient power generation and distribution systems that perform reliably.

As mentioned before, design teams use robust design methodologies to manage complex energy generation and distribution problems, including an alternator charging system or taking into account system and environmental variations that affect performance. Saber's comprehensive simulation, modelling resources and analysis capabilities enable design teams to meet the challenges of powertrain design.

Advantages for design engineers in vehicle power network. Saber allows design engineers to accurately size system components so as to match the

energy generation with corresponding consumption. It eliminates untoward surprises in power network loading and distribution through early detection and verification. "Saber also allows early testing of the complete system with hardware/software co-simulation." It saves time and eliminates errors with industry-proven power model libraries apart from enabling design portability using industry-standard VHDL-AMS and MAST languages. The Saber platform maximises reliability with advanced sensitivity, statistical performance and fault analyses while increasing analysis throughput with distributed simulations across multiple CPUs.

In-vehicle networks (IVNs). Advances in automotive design have been propelled by a dramatic increase in electronics content. Today, a whopping one million lines of code go into a car's electronics. Numerous electronic systems are required to communicate with each other over a complex network of different communication protocols, such as CAN, LIN and FlexRay. These complexities add significant challenges to physical network layer verification for seamless communication.

With a traditional approach to prototyping, it is nearly impossible to build enough prototypes to adequately test the most important and critical variations of a network design. This is where virtual prototyping comes as the solution for verifying data-network reliability. Saber provides a compre-

System Requirements

OS: Windows XP SP3 or higher version
CPU: Intel Atom D525 @ 1.8 GHz
RAM: 2GB DDR 3
HDD: At least 1.5 GB free space

hensive simulation and analysis capability – something which is extremely necessary to apply robust design methodologies to in-vehicle networks and ensure reliability.

Advantages of using Saber for in-vehicle network. Saber makes it easy to verify network concepts and topologies early in the development cycle, thus sparing designers the cumbersome task of integrating code later. Using Saber, it is possible to analyse specific network variants (for example, minimum vs maximum number of electronic control units), which saves more time while eliminating errors with the help of industry-proven IVN libraries. In order to analyse possible topology extensions, Saber comes with the ability to include wire characteristics to the design. This feature increases robustness while increasing analysis throughput.

Powertrain and hybrid-electric vehicle systems. Consumer demand for higher fuel efficiency and lower vehicle emissions has accelerated the development of pure electric (EV) and hybrid electric (HEV) drive-trains. These vehicles depend on advanced electronically-controlled systems working together across a wide range of operating conditions to ensure efficient performance, safety and reliability. Increasing electrical content and complexity, coupled with shorter design cycles, require design teams to continually improve their design methods for mechatronic integration. To minimise risk and potential recall or re-design, engineers rely on simulation-based robust design methods to deliver predictable and reliable designs. Saber's comprehensive modeling, simulation and analysis capabilities are applied to improve system performance, reduce cost and maximise reliability of these critical systems.

Saber advantages in HEV/EV systems. Design engineers can use Saber